DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY (DU)

M.Tech /MS/Ph.D EXAMINATIONS

MID SEMESTER- III (18/05/2023)

Course (Subject) : **Simulation of Linear and Nonlinear Systems** Course Code : **AM-622**

Duration : **01 hr** Max Marks : **10**

Note: 1. Marks are indicated against each question.

2. Attempt **both** questions from the following .

1. a). Explain briefly about "Classification of Queue" with usual notations and various models.

b). For the queue model (M / M / 1) : (∞ / FIFO) , with the usual notation derive average length of non-empty queue [2+3 = 5 M]

2 a). In a railway marshalling yard, goods trains arrive at a rate of 30 trains per day. Assuming that the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with an average 36 minutes. Calculate the following:

i). The probability that the yard is empty. ii). The average queue length.

Assuming that the capacity of the yard is 5 trains only.

[3+2 = 5 M]

b). Discuss about the stability of non-linear systems

M.Tech /MS/Ph.D EXAMINATIONS

END SEMESTER: May/June- 2023

Course (Subject) : **Simulation of Linear and Nonlinear Systems** Course Code: **AM-622**

Duration : **03 hrs** Max Marks : **50**

Note:

1. Marks are indicated against each question.
2. Write clearly and legibly. All diagrams to be neatly labelled.
3. Before attempting, please ensure that you have the correct question paper.
4. Attempt **all five questions**.
5. All questions carry equal marks.
6. **Calculator is permitted & students may request the invigilator for required Tables.**

1).a). Explain briefly about i). Simulation ii). Randam Variate iii). Random Number iv). Pseudo-random numbers.

b). Using the multiplicative congruential method, find the Period of the generator for $a = 13$, $m = 2^6 = 64$ and $X_0 = 1, 2, 3$. Hence write your observations. **(4 + 6 = 10 M)**

2). a). The probability density function of a continuous random variable is given by $f(x) = \sin x$ in $[0, \frac{\pi}{2}]$ then find Mean, Variance and Mode.

b). Write the general procedure to generate random variates.

c). Find the random variates X_i corresponding to the random numbers $R_i = 0.78, 0.03, 0.23, 0.97$ using discrete uniform distribution over $\{1,2,3,4,5,6,7,8,9,10\}$. (4+2+4=10 marks)

3). a). For the queue model (M / M / 1) : (∞ / FIFO) , with the usual notation find

(i). Average no. of customers in the system E(n) and

(ii). Average waiting time that an arrival spends in the system W_s (or) E(v).

b).In a super market the average arrival rate of customers is 5 for every 30 minutes. The average time takes to list and calculate the customers purchases at the cast desk is 4.5 minutes and this time is exponentially distributed.

i). How long will be customer expect to wait for the service at cash desk.

ii). What is the choice the queue length will exceed 5.

iii). What is the probability that the cashier is working. (5+5 = 10 marks)

4). Use the Chi-square test with $\alpha = 0.05$ to test for whether the data shown below is uniformly distributed or not. (10 marks)

0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
0.96	0.99	0.77	0.67	0.56	0.41	0.52	0.73	0.99	0.02
0.47	0.30	0.17	0.82	0.56	0.05	0.45	0.31	0.78	0.05
0.79	0.71	0.23	0.19	0.82	0.93	0.65	0.37	0.39	0.42
0.99	0.17	0.99	0.46	0.05	0.66	0.10	0.42	0.18	0.49
0.37	0.51	0.54	0.01	0.81	0.28	0.69	0.34	0.75	0.49
0.72	0.43	0.56	0.97	0.30	0.94	0.96	0.58	0.73	0.05
0.06	0.39	0.84	0.24	0.40	0.64	0.40	0.19	0.79	0.62
0.18	0.26	0.97	0.88	0.64	0.47	0.60	0.11	0.29	0.78

5). a). Define Linear and non-Linear systems and write their general properties. **(3 + 7 = 10 M)**

b). Explain briefly about i). Lyapunov Stability ii). Uniform stability, iii). Asymptotic stability and iv). Exponential stability

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Assessment-I- 18/03/2024

Course (Subject) : **Simulation of Linear and non-Linear systems** Course Code : **AM-322**

Duration : **01hrs** Max Marks : **10**

1. a). Define uniform probability distribution. Hence find mean and variance.

   b). Define i). Random Variate ii). Random Number

2. a). Explain about "Testing of Hypothesis".

   b) Use the Kolmogoro – Simulation test with $\alpha = 0.05$ to learn whether the sequence of random numbers 0.54, 0.73, 0.98, 0.11 & 0.68.are uniformly distributed or not. [N=5, $\alpha = 0.05$, D = 0.565]

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M.Tech /MS/Ph.D EXAMINATIONS

MID SEMESTER- III (22/04/2024)

Course (Subject) : **Simulation of Linear and Nonlinear Systems** Course Code : **AM-622**

Duration : **01 hr** Max Marks : **10**

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1 . a). Write the general procedure to generate the sequence of random variates.

b). Find the random variates X_i corresponding to the random numbers R_i = 0.78, 0.03, 0.23, 0.97 using discrete uniform distribution over $\{1,2,3,4,5,6,7,8,9,10\}$.

c). If $f(x)=\frac{x^2}{9}$, $0 \le x \le 3$ is probability density function, then derive an equation to generate the random variate X_i . Also find its mean and Variance.

4). a). For the queue model (M / M / 1) : (∞ / FIFO) , with the usual notation find

(i). Average length of non-empty queue

(ii). Variance of the above model

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M.Tech /MS/Ph.D EXAMINATIONS

MID EXAM- III (10/05/2024)

Course (Subject) : **AM -622- Simulation of Linear and Nonlinear Systems** Max Marks : **10**

Note: Attempt **all** the three questions from the following each question carries 2.5 marks.

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1). Explain briefly the applications of Queuing theory.

2). Discuss about the stability of non-linear systems

3). Write a short note on Limit cycles and Bifurcations.

4). Write a short note on LYAPUNOV function and its applications.

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY (DU)

M.Tech / Ph.D END SEMESTER Exams-May 2024

Course : **Simulation of Linear and Non Linear systems** Course Code : AM-622

Duration : 03 hrs Max Marks : 50

Note:

1. Before attempting ,please ensure that you have the correct question paper.
2. Attempt **All Five** questions from the following. All questions carry equal marks.
3. Caluculator is permitted & students may request the invigilator for required Tables.

1). a). Explain briefly about Pseudo-random numbers.

b). Using the multiplicative congruential method, find the Period of the generator for

$a = 13$, $m = 2^6 = 64$ and $X_0 = 1, 2, 3, 4$. Hence write your observations. **(3 + 7= 10 M)**

2). a). For the queue model (M / M / 1) : (∞ / FIFO), with the usual notation find

(i). Average no. of customers in the system and

(ii). Average waiting time of an arrival **(5+5=10 M)**

b). A barber shop has space to accommodate only 10 customers. He can serve only one person at a time. If a customer comes to his shop and finds with full, he goes to the next shop. Customers randomly arrive at an average rate $\lambda = 10$ per hour and the barber service time is negative exponential with an average of $\frac{1}{\mu} = 5$ minutes / customer the find P_0 , P_n and E(n).

3). a). Derive the expression for Random Variate X_i for the Triangular distribution with probability density function as

$$f(x) = \begin{cases} x, 0 < x \leq 1 \\ 2 - x, 1 < x \leq 2 \\ 0, otherwise \end{cases}$$

b). Find the random variates X_i for the random numbers $R_i = 0.1306, 0.0422, 0.6597, 0.7965$ Using exponential distribution with mean=1 **(5+5=10 M)**

4). Use the Chi-square test with $\alpha = 0.05$ to test for whether the data shown below is uniformly distributed or not. [Given, Chi-Square with degree freedom 9, α = 5% is 16.92) **[10 Marks]**

0.34	0.90	0.25	0.89	0.87	0.44	0.12	0.21	0.46	0.67
0.83	0.76	0.79	0.64	0.70	0.81	0.94	0.74	0.22	0.74
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0.99	0.17	0.99	0.46	0.05	0.66	0.10	0.42	0.18	0.49
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0.06	0.39	0.84	0.24	0.40	0.64	0.40	0.19	0.79	0.62
0.18	0.26	0.97	0.88	0.64	0.47	0.60	0.11	0.29	0.78

5). a). **Define Linear and non-Linear systems and write their general properties.** **(5 + 5 = 10 M)**

b). **Explain briefly about i). Lyapunov Stability ii). Uniform stability, iii). Asymptotic stability and iv). Exponential stability**

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